IV. EDA-Data Inspection and Analysis

AIM:

- Viewing and inspecting DataFrames
- Filtering and subsetting data using conditions
- Descriptive statistics: measures of central tendency (mean, median, mode) and measures of dispersion (range, variance, standard deviation)

PROCEDURE:

1. Import the required Python libraries, such as Pandas and NumPy, for data analysis.
2. Load the dataset into a Pandas DataFrame using pd.read_csv().
3. View and inspect the DataFrame using head(), tail(), and sample() to examine the records.
4. Check the structure and information of the dataset using shape, columns, dtypes, and info().
5. Filter rows based on specific conditions using comparison operators and Boolean indexing.
6. Select and extract the required rows and columns using loc[], iloc[], and column names.
7. Calculate measures of central tendency, such as mean, median, and mode, using mean(), median(), and mode().
8. Calculate measures of dispersion, such as range, variance, and standard deviation, using max() - min(), var(), and std().
9. Display and interpret the filtered data and descriptive statistics to understand the distribution and variability of the dataset.

PROGRAM:

```
import pandas as pd

df = pd.read_csv('/content/test_Y3wMUE5_7gLdaTN.csv')

print(df.head())

print(df.tail())
```

```
print(df.info())

print(df.describe())

print(df.columns)
```

```
366  LP002989   Male      No          0      Graduate           Yes

     ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term  \
362             4009               1777       113.0             360.0
363             4158                709       115.0             360.0
364             3250               1993       126.0             360.0
365             5000               2393       158.0             360.0
366             9200                  0        98.0             180.0

     Credit_History Property_Area
362             1.0         Urban
363             1.0         Urban
364             NaN     Semiurban
365             1.0         Rural
366             1.0         Rural
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 367 entries, 0 to 366
Data columns (total 12 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Loan_ID            367 non-null    object
 1   Gender             356 non-null    object
 2   Married            367 non-null    object
 3   Dependents         357 non-null    object
 4   Education          367 non-null    object
 5   Self_Employed      344 non-null    object
 6   ApplicantIncome    367 non-null    int64
 7   CoapplicantIncome  367 non-null    int64
 8   LoanAmount         362 non-null    float64
 9   Loan_Amount_Term   361 non-null    float64
 10  Credit_History     338 non-null    float64
 11  Property_Area      367 non-null    object
dtypes: float64(3), int64(2), object(7)
memory usage: 34.5+ KB
None
```

```
categorical_cols = ['Gender', 'Married', 'Dependents', 'Education',
'Self_Employed', 'Property_Area']


for col in categorical_cols:
    unique_vals = df[col].unique()
    print(f"\nUnique values in '{col}': {unique_vals}")
```

```
Unique values in 'Gender': ['Male' 'Female' nan]

Unique values in 'Married': ['Yes' 'No']

Unique values in 'Dependents': ['0' '1' '2' '3+' nan]

Unique values in 'Education': ['Graduate' 'Not Graduate']

Unique values in 'Self_Employed': ['No' 'Yes' nan]

Unique values in 'Property_Area': ['Urban' 'Semiurban' 'Rural']
```

```
df['Dependents'] = df['Dependents'].replace('3+', 3)

df['Dependents'] = pd.to_numeric(df['Dependents'],
errors='coerce').astype('Int64')

print("\nValue counts in 'Dependents' after conversion:")
```

```
print(df['Dependents'].value_counts(dropna=False))
```

```
Value counts in 'Dependents' after conversion:
Dependents
0       200
2        59
1        58
3        40
<NA>     10
Name: count, dtype: Int64
```

```
cols = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']


for col in cols:
    print(f"\nStatistics for '{col}':")
    print(f"Mean: {df[col].mean():.2f}")
    print(f"Median: {df[col].median():.2f}")
    print(f"Mode: {df[col].mode().values}")
    print(f"Range: {df[col].max() - df[col].min():.2f}")
    print(f"Variance: {df[col].var():.2f}")
    print(f"Standard Deviation: {df[col].std():.2f}")
```

```
Statistics for 'ApplicantIncome':
Mean: 4805.60
Median: 3786.00
Mode: [3500 5000]
Range: 72529.00
Variance: 24114831.09
Standard Deviation: 4910.69

Statistics for 'CoapplicantIncome':
Mean: 1569.58
Median: 1025.00
Mode: [0]
Range: 24000.00
Variance: 5448639.49
Standard Deviation: 2334.23

Statistics for 'LoanAmount':
Mean: 136.13
Median: 125.00
Mode: [150.]
Range: 522.00
Variance: 3765.87
Standard Deviation: 61.37
```

RESULT:

Thus, the given program was written and executed successfully.